API Design & Data Flow

1. Overview

The API layer connects the frontend of BuyBlink with the backend server. It allows secure communication between the user interface and the database through RESTful endpoints. All data exchanges are performed using JSON format.

2. API Endpoints

◆ **Authentication APIs**

1. Register User

- Method: POST
- Endpoint: /api/auth/register
- Request Body:

{
  "name": "John",
  "email": "john@email.com",
  "password": "123456"
}

- Response:

{
  "message": "User registered successfully"
}

2. Login User

- Method: POST
- Endpoint: /api/auth/login
- Request Body:

{
  "email": "john@email.com",
  "password": "123456"
}

- Response:

{
  "token": "jwt_token_here",
  "role": "user"
}

🔹 Product APIs

3. Get All Products

- Method: GET
- Endpoint: /api/products
- Response:

[
  {
    "product_id": 1,
    "name": "Eco Bottle",
    "price": 299,
    "eco_rating": 5
  }
]

4. Add Product (Admin Only)

- Method: POST
- Endpoint: /api/products
- Request Body:

{
  "name": "Eco Bag",
  "price": 199,
  "stock": 50,
  "category_id": 2
}

◆ Order APIs

5. Create Order

- Method: POST
- Endpoint: /api/orders
- Request Body:

```
{
 "user_id": 1,
 "items": [
  { "product_id": 1, "quantity": 2 }
 ]
}
```

6. Get User Orders

- Method: GET
- Endpoint: /api/orders/user/:id

3. Frontend–Backend Data Flow

1. User interacts with frontend (React / HTML UI)
2. Frontend sends HTTP request to backend API
3. Backend processes request
4. Backend fetches data from database
5. Backend returns JSON response
6. Frontend displays updated data

4. Technology Used for API

- REST Architecture
- JSON Data Format
- HTTP Methods (GET, POST, PUT, DELETE)
- JWT Authentication (optional)

5. Conclusion

The API structure ensures modular communication between frontend, backend, and database, enabling scalable and secure operations within the BuyBlink platform.